U3: The Forced Structure of Time-Translation Symmetry and Schrödinger Evolution

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Abstract

We present the structural derivation of the participation-measure evolution equation $i\hbar\,\partial_t P_K = \hat{H} P_K$ — the Schrödinger equation in standard non-relativistic single-particle form — from Primitive 13 (relational timing), Primitive 03 (graph homogeneity), and the now-FORCED upstream items U1, U2-Discrete, U2-Continuum, U5, and U4. Operating under a strict identification-not-derivation discipline (U4 may appear in the F5 step only as the identification target, never as a derivation premise), we decompose U3 into five sub-features: existence of a strongly continuous time-translation representation on the U2 Hilbert space (F1); uniqueness of its self-adjoint generator via Stone's theorem (F2); linearity of the induced evolution equation (F3); first-order-in-time structure (F4); and compatibility of the Stone-derived generator $\hat{H}_{U3}$ with the U4-derived kinetic-plus-potential operator $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ (F5). F1 follows from Primitive 13 (continuous time axis) plus Primitive 03 (homogeneity) plus U2 inner-product translation-invariance. F2 follows from Stone's theorem on one-parameter unitary groups. F3 and F4 are automatic from Hilbert-space structure and the operator-exponential parameterization. F5 — the load-bearing content — is established by **Galilean Lie algebra closure**: $\hat{H}_{U3}$, as the time-translation generator within the Galilean representation on the U2 Hilbert space, must satisfy the commutator $[\hat{H}_{U3}, \hat{K}_i] = -i\hbar\hat{p}_i$ with the boost generator $\hat{K}_i = m\hat{x}_i - t\hat{p}_i$; integrating against the U5 momentum operator yields $T(\hat{p}) = |\hat{p}|^2/(2m)$ via the chain-rule Jacobian, identifying $\hat{H}_{U3}$ with $\hat{H}_{U4}$ modulo the additive vacuum-energy zero-point gauge. The identification is cross-checked by Stone-theorem uniqueness applied to the $\hat{H}_{U4}$-generated unitary group. All seven falsifiers are dispatched. The U3 verdict is **forced**, with no new structural CANDIDATEs introduced and zero new conditionalities. **U4 is retroactively promoted from forced-conditional to forced-unconditional**, resolving its Framing-1 sibling-CANDIDATE conditionality. The active upstream-CANDIDATE inventory of the QM-emergence Phase-1 program reduces from $\{U3\}$ + continuum gauge to $\{\,\}$ + continuum gauge — empty modulo description-level convention. **All four foundational postulates of non-relativistic single-particle quantum mechanics — the Born rule, the Bell–Tsirelson bound, the Heisenberg uncertainty inequality, and the Schrödinger evolution equation — are now structurally forced-unconditional consequences of the Event Density primitive ontology.** The forced-theorem inventory rises to sixteen with two retroactive promotions; Phase-1 of the QM-emergence program is structurally complete.

1 Introduction

1.1 What U3 is

U3 is the structural commitment that the participation-measure components P_K on the U2 Hilbert space $\mathcal{H}$ evolve in time according to a single-generator linear first-order equation,

$$i\hbar\,\partial_t P_K(x,t) = \hat{H} P_K(x,t),$$

with $\hat{H}$ a self-adjoint operator on $\mathcal{H}$. Combined with U4's specific Hamiltonian form $\hat{H} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$, the equation is the standard Schrödinger equation of non-relativistic single-particle quantum mechanics. U3 supplies the existence, uniqueness, linearity, and first-order character of the equation; U4 supplies the form of the generator $\hat{H}$.

The earlier framing of U3 in the QM-emergence Phase-1 inventory bundled it with U4 as a single dynamical commitment. The present arc separates the two cleanly: U4 was closed first (in the 2026-04-26 cycle, conditional on U3 via Framing 1); U3 is now closed independently (this paper), retroactively resolving U4's conditionality. The two-arc separation is methodologically essential to preserve acyclicity in the U3 $\leftrightarrow$ U4 loop.

1.2 Why U3 matters

U3 is **the final arc of the QM-emergence Phase-1 structural-foundations program**. With U3 closed, every upstream commitment in the participation-measure $\rightarrow$ Hilbert-space $\rightarrow$ Schrödinger-emergence chain has been derived from the ED primitive ontology plus standard mathematical and symmetry-group infrastructure. The four foundational postulates of non-relativistic single-particle quantum mechanics — the Born rule (Theorem 10), the Bell–Tsirelson bound, the Heisenberg uncertainty inequality, and the Schrödinger evolution equation — are now **forced-unconditional structural consequences of ED primitives**.

The result is the closure of a seven-arc cycle (born_gleason $\rightarrow$ U2-Discrete $\rightarrow$ U2-Continuum $\rightarrow$ U5 $\rightarrow$ U1 $\rightarrow$ U4 $\rightarrow$ U3) in which the structural-derivation methodology has been demonstrated against seven substantively different structural questions. Phase-1 is structurally complete, modulo the description-level continuum gauge convention established in U2-Continuum.

1.3 Relationship to U1, U2, U5, U4

- **U1.** U3 takes the participation-measure construction $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$ as input. The $U(1)$-valued polarity-phase factor supplies the complex structure on which Hermitian operators act and on which time evolution preserves the inner product.
- **U2.** U3 takes the U2 Hilbert space $\mathcal{H}$ and its sesquilinear inner product as input. Time-translation operators are unitary on $\mathcal{H}$; Stone's theorem applies.
- **U5.** U3 takes the U5 momentum operator $\hat{p} = -i\hbar\nabla$ as input for the F5 Galilean-closure step. The U5 derivation also serves as the methodological template for the U3 arc: Stone's theorem applied to time-translation symmetry mirrors the U5 application to spatial-translation symmetry.
- **U4.** U3 treats U4's operator form $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ as the **identification target** in F5, never as a derivation premise. The strict discipline that distinguishes "U4 as identification target" from "U4 as derivation input" preserves acyclicity; without this discipline, U3's derivation would invert U4's Framing-1 conditionality and close a circular loop.

1.4 Summary of the U3 verdict

The U3 verdict is **forced**. The five sub-features F1, F2, F3, F4, F5 close uniformly under the available structural inputs and the identification-not-derivation discipline. All seven falsifiers (Fal-1 through Fal-7) dispatch. No new structural CANDIDATEs are introduced. **U4 is retroactively promoted to forced-unconditional**, completing the structural derivation of the Schrödinger equation.

The downstream cascade is decisive:

- The active upstream-CANDIDATE inventory reduces from $\{U3\}$ + continuum gauge to $\{\,\}$ + continuum gauge.
- The forced-theorem count rises from fifteen to sixteen, with two retroactive promotions in this cycle (U3 from CANDIDATE to FORCED; U4 from FORCED-conditional to FORCED-unconditional).
- All four foundational QM postulates are FORCED-unconditional.
- **Phase-1 of the QM-emergence program is structurally complete** (modulo description-level continuum gauge).

1.5 Organization

Section 2 states the U3 question precisely and inventories its sub-features and falsifiers. Section 3 inventories the structural inputs and the inputs explicitly forbidden under the identification-not-derivation discipline. Section 4 decomposes U3 into F1–F5 with status assignments. Section 5 introduces the seven falsifiers. Sections 6–9 derive F1, F2, F3, F4 in turn. Section 10 derives the load-bearing F5 via Galilean Lie algebra closure (Strategy A) with Stone-uniqueness as cross-check (Strategy B). Section 11 consolidates the falsifier dispatches. Section 12 documents the conditionality ledger. Section 13 states the final U3 theorem. Section 14 documents the downstream cascade including Phase-1 completion. Section 15 discusses the result's significance and identifies the next-priority work.

2 Statement of the U3 Question

2.1 The target structure

The target structural content of U3 is the existence, uniqueness, linearity, and first-order-in-time character of the participation-measure evolution equation, together with the identification of its generator with the U4-derived Hamiltonian operator.

Formally: let $\mathcal{H}$ denote the U2 Hilbert space of participation measures (discrete or continuum regime, gauge-class equivalent under the U2-Continuum conformal redundancy). Let U_t for $t \in \mathbb{R}$ denote the time-translation operator $(U_t P)_K(x, s) := P_K(x, s - t)$ acting componentwise on participation-measure components. The U3 question is whether $\{U_t : t \in \mathbb{R}\}$ is a strongly continuous one-parameter unitary group whose unique self-adjoint generator $\hat{H}_{U3}$ coincides with the U4-derived operator $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$, yielding the Schrödinger evolution equation

$$i\hbar\,\partial_t P(t) = \hat{H}P(t), \qquad \hat{H} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x}). \tag{1}$$

2.2 The five sub-features

- **F1 (Existence of strongly continuous time-translation representation).** $\{U_t : t \in \mathbb{R}\}$ is a well-defined, strongly continuous, one-parameter abelian unitary group on $\mathcal{H}$.
- **F2 (Uniqueness of the self-adjoint generator).** There exists a unique self-adjoint operator $\hat{H}_{U3}$ on $\mathcal{H}$ such that $U_t = \exp(-i\hat{H}_{U3}t/\hbar)$.
- **F3 (Linearity of the evolution equation).** The induced equation $i\hbar\,\partial_t P = \hat{H}_{U3}P$ is linear in P.
- **F4 (First-order-in-time structure).** The evolution equation is first-order in ∂_t.
- **F5 (Compatibility with U4 Hamiltonian form).** The Stone-derived generator $\hat{H}_{U3}$ coincides as an operator with the U4-derived $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$.

2.3 The seven falsifiers

- **Fal-1 (Non-unitary time evolution).**
- **Fal-2 (Multiple inequivalent generators).**
- **Fal-3 (Non-linear evolution).**
- **Fal-4 (Higher-order-in-time evolution).**
- **Fal-5 (Generator incompatible with U4).**
- **Fal-6 (Non-self-adjoint generator).**
- **Fal-7 (Non-continuous time-translation representation).**

2.4 Methodological framing: identification, not derivation

The U3 arc adopts a strict discipline distinguishing two roles for U4:

- **Forbidden:** U4 as a derivation premise. Using U4's operator form to derive $\hat{H}_{U3}$ would invert U4's Framing-1 conditionality (U4 was derived conditional on U3's existence-of-Hermitian-generator content) and close a circular loop.
- **Permitted:** U4 as the identification target in F5. The F5 step *identifies* $\hat{H}_{U3}$ with $\hat{H}_{U4}$ via the shared Galilean Lie algebra structure — both operators land on the same Hamiltonian because the Galilean group's representation theory forces them to, not because either is derived from the other.

This protocol preserves acyclicity in the U3 $\leftrightarrow$ U4 loop. The discipline is enforced by an explicit circularity audit (§3.4).

3 Structural Inputs

3.1 Primitives

- **Primitive 13 (Relational Timing).** Supplies the continuous time axis $\mathbb{R}_t$ and time-translation symmetry as a kinematic feature of the participation graph. Central new input of this arc.
- **Primitive 03 (Participation).** Graph homogeneity, including time-axis homogeneity (no privileged instant of time).
- **Primitive 04 (Bandwidth).** Supplies the participation-measure components P_K on which U_t acts.
- **Primitive 09 (Tension Polarity).** $U(1)$-valued polarity phase. Supplies the complex structure on which Hermitian operators act (via U1).

3.2 Inherited forced upstream items

- **U1 (Theorem 14).** Participation-measure construction $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$.
- **U2-Discrete (Theorem 11) and U2-Continuum (Theorem 12).** Sesquilinear inner product on the participation-measure Hilbert space, in both regimes.
- **U5 (Theorem 13).** Spatial translation generator $\hat{p} = -i\hbar\nabla$. Methodological template; supplies the momentum operator entering F5.

- **U4 (Theorem 15).** Specific Hamiltonian form $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$. **Used solely in F5 as the identification target.**
- **Theorem 10 (Born rule, Gleason–Busch).** Available as background.

3.3 Mathematical infrastructure

- Stone's theorem on one-parameter unitary groups (Stone 1932) [9].
- Spectral theorem for self-adjoint operators on Hilbert spaces.
- Standard L^2 functional analysis: translation continuity, operator-exponential calculus [11].
- Galilean Lie algebra and central extension (Bargmann 1954) [10].

3.4 Forbidden inputs (circularity audit)

The following inputs are explicitly forbidden as derivation inputs to U3:

- **U4 as derivation premise** for F1, F2, F3, F4. U4 may appear in F5 only as identification target, never as a derivation input.
- **U4's Framing-1 "U3 exists" assumption.** Forbidden as input to F5; F5 treats U4's operator form as a mathematical object on $\mathcal{H}$, not as a conditional construct.
- **U3 itself** (question-begging). Forbidden in any role.
- **The Schrödinger equation in textbook form.** Forbidden as derivation input.
- **Relativistic kinematics.** Out of scope.
- **Downstream gauge-coupling structure.** Out of scope (gauge-coupling-free per inherited Phase-1 Step 2 scope).
- **QFT-level inputs.** Out of scope.

Each subsequent derivation operates strictly within the primitive-input scope of the present arc. The circularity audit follows the canonical templates of U1 Memo 01 §4, U4 Memo 01 §2, and U5 §3.5.

3.5 Scope conditions

- **Non-relativistic single-particle scope.** Inherited from Phase-1 Step 2; selects the abelian $\mathbb{R}_t$ time-translation group and the Galilean Lie algebra.
- **Gauge-coupling-free scope.** Inherited from U4.
- **Flat-spacetime baseline (continuum regime).**
- **Continuous-time interpretation of Primitive 13.** Confirmed by pre-derivation audit at Phase-1 scope.

4 Sub-Feature Decomposition

Sub-feature	Status	Structural inputs	Falsifiers
F1 (translation rep.)	FORCED	P-13 + P-03 + U2 inner-product translation-invariance	Fal-1, Fal-7
F2 (unique generator)	FORCED	F1 + Stone's theorem	Fal-2, Fal-6
F3 (linearity)	automatic	F1 + F2 + Hilbert-space operator structure	Fal-3
F4 (first-order-in-t)	automatic	F2 + non-rel scope	Fal-4
F5 (compat. with U4)	FORCED (load-bearing)	F1 + F2 + U5 momentum + Galilean Lie algebra; Stone-uniqueness cross-check	Fal-5

Two forced-via-derivation items (F1, F2), two automatic items (F3, F4), one load-bearing item (F5).

5 Falsifier Inventory

- **Fal-1 (Non-unitary time evolution).** U_t fails to preserve the U2 inner product. Resolved §6.4.
- **Fal-2 (Multiple inequivalent generators).** Stone's theorem fails. Resolved §7.2.
- **Fal-3 (Non-linear evolution).** Resolved §8.2.
- **Fal-4 (Higher-order-in-time).** Klein–Gordon-like second-order equation. Resolved §9.2.
- **Fal-5 (Generator incompatible with U4).** $\hat{H}_{U3} \neq \hat{H}_{U4}$. Resolved §10.
- **Fal-6 (Non-self-adjoint generator).** Resolved §7.3.
- **Fal-7 (Non-continuous time-translation representation).** Resolved §6.6.

6 Derivation of F1 (Time-Translation Representation)

6.1 The claim

The time-translation operators $\{U_t : t \in \mathbb{R}\}$ acting on $\mathcal{H}$ by

$$(U_t P)_K(x, s) := P_K(x, s - t)$$

form a strongly continuous one-parameter abelian unitary group.

6.2 Time-translation symmetry as a kinematic feature

Three primitive-level observations:

(i) Primitive 13 supplies a continuous time axis $\mathbb{R}_t$ at the structural level.

(ii) Primitive 03's homogeneity extends to the time axis: the participation graph's structure at time s is structurally identical to its structure at time $s - t$, up to relabeling.

(iii) No primitive-level structure singles out a privileged time origin or magnitude.

These observations establish time-translation symmetry as a *kinematic* property of the participation graph. **No reference to U4, no Hamiltonian, and no Schrödinger equation appears in this argument.**

6.3 Linearity

Time translation acts linearly on participation-measure components:

$$U_t(\alpha P + \beta Q)_K(x, s) = \alpha(U_t P)_K(x, s) + \beta(U_t Q)_K(x, s)$$

for all $\alpha, \beta \in \mathbb{C}$ and $P, Q \in \mathcal{H}$.

6.4 Unitarity

For $P, Q \in \mathcal{H}$ and $t \in \mathbb{R}$:

$$\langle U_t P \mid U_t Q \rangle_{\mathcal{H}} = \sum_K \int_{\mathbb{R}^d} \int_{\mathbb{R}_t} P_K^*(x, s-t)\, Q_K(x, s-t)\, ds\, d\mu(x).$$

Substituting $s' = s - t$, with $ds' = ds$ (the U2 measure is translation-invariant on the time axis: Primitive 13 supplies translation-invariant relational time, with no t-dependent weighting):

$$\langle U_t P \mid U_t Q \rangle_{\mathcal{H}} = \langle P \mid Q \rangle_{\mathcal{H}}.$$

Falsifier Fal-1 dispatched.

6.5 Group structure

Direct calculation: $U_t U_s = U_{t+s}$, $U_0 = I$, $U_{-t} = U_t^{-1}$. Abelian one-parameter group.

6.6 Strong continuity

For any $P \in \mathcal{H}$ with components $P_K(x, \cdot) \in L^2(\mathbb{R}_t, ds)$, the map $t \mapsto U_t P$ is strongly continuous because L^2-translation is strongly continuous (standard result of L^2 functional analysis [11]). Continuity inherits to the U2 Hilbert space's direct-integral structure.

The continuous-time interpretation of Primitive 13 is the precondition. Pre-derivation audit confirms continuous time at Phase-1 scope. **Falsifier Fal-7 dispatched.**

Theorem 6.1 (F1 — Time-Translation Representation)**.** *Let $\mathcal{H}$ denote the U2 Hilbert space of participation measures. Then the time-translation operators $(U_t P)_K(x, s) := P_K(x, s-t)$ for $t \in \mathbb{R}$ form a strongly continuous one-parameter abelian unitary group on $\mathcal{H}$. The derivation invokes Primitives 13 and 03 plus the U2 inner-product structure, with no input from U4, no Hamiltonian, and no PDE-level Schrödinger assumption.*

7 Derivation of F2 (Stone-Uniqueness)

7.1 Stone's theorem

Theorem 7.1 (Stone 1932 [9])**.** *Every strongly continuous one-parameter unitary group $\{V(t) : t \in \mathbb{R}\}$ on a Hilbert space has a unique self-adjoint generator A such that $V(t) = \exp(itA)$.*

7.2 Application to F1's group

By Theorem F1, $\{U_t : t \in \mathbb{R}\}$ is a strongly continuous one-parameter abelian unitary group on $\mathcal{H}$. Apply Stone's theorem: there exists a unique self-adjoint operator $\hat{H}_{U3}$ on $\mathcal{H}$ such that

$$U_t = \exp\left(-\frac{i\hat{H}_{U3}t}{\hbar}\right). \tag{2}$$

The generator is unique. **Falsifier Fal-2 dispatched.**

7.3 Self-adjointness

Stone's theorem produces a self-adjoint generator from a strongly continuous unitary group. Self-adjointness is automatic. **Falsifier Fal-6 dispatched.**

7.4 No U4-dependent assumptions enter

The derivation invokes only F1 and Stone's theorem. U4's specific operator form does not enter. The operator $\hat{H}_{U3}$ is *abstractly* the time-translation generator; identification with U4's operator is the F5 question.

Theorem 7.2 (F2 — Uniqueness of the Time-Translation Generator)**.** *Under the hypotheses of Theorem F1, there exists a unique self-adjoint operator $\hat{H}_{U3}$ on $\mathcal{H}$ such that $U_t = \exp(-i\hat{H}_{U3}t/\hbar)$ for all $t \in \mathbb{R}$. The identification is U4-independent.*

8 Derivation of F3 (Linearity)

8.1 The argument

Differentiating the operator-exponential at $t = 0$:

$$\left.\frac{d}{dt}\right|_{t=0} U_t P = -\frac{i}{\hbar}\hat{H}_{U3}P.$$

For the time-evolved state $P(t) := U_t P_0$:

$$i\hbar\,\partial_t P(t) = \hat{H}_{U3}P(t). \tag{3}$$

8.2 Linearity is automatic

Equation (3) is linear in $P(t)$ because: (i) $\mathcal{H}$ is a complex vector space; (ii) $\hat{H}_{U3}$ is a self-adjoint operator on $\mathcal{H}$, hence linear; (iii) $\{U_t\}$ acts linearly on $\mathcal{H}$ (per F1); (iv) time translations form an additive abelian group with $U_{t+s} = U_t U_s$ inheriting linearity.

Gross–Pitaevskii-like nonlinearities are phenomenological; they do not arise structurally in the Phase-1 single-particle scope. **Falsifier Fal-3 dispatched.**

Theorem 8.1 (F3 — Linearity)**.** *The induced evolution equation $i\hbar\,\partial_t P(t) = \hat{H}_{U3}P(t)$ is linear in $P(t)$. Linearity is automatic from the Hilbert-space structure, the operator-theoretic definition of $\hat{H}_{U3}$, the linearity of the unitary representation $\{U_t\}$, and the additive structure of the time-translation group.*

9 Derivation of F4 (First-Order-in-Time)

9.1 The argument

Differentiating $U_t = \exp(-i\hat{H}_{U3}t/\hbar)$ once in t produces a first-order equation (3).

9.2 Why higher-order does not arise

Higher derivatives produce powers of $\hat{H}_{U3}$:

$$i\hbar\,\partial_t^n P(t) = (\hat{H}_{U3})^n P(t),$$

which are downstream consequences of (3), not independent equations.

A second-order-in-time equation (Klein–Gordon-like) arises in *relativistic* quantum mechanics. The non-relativistic scope condition (inherited from Phase-1 Step 2's framing, not from U4) selects the abelian $\mathbb{R}_t$ time-translation group; under this scope, the operator-exponential structure gives a first-order equation. **Falsifier Fal-4 dispatched.**

Theorem 9.1 (F4 — First-Order-in-Time)**.** *The evolution equation $i\hbar\,\partial_t P(t) = \hat{H}_{U3}P(t)$ is first-order in ∂_t. The first-order character is automatic from Stone's theorem, the abelian group structure of time translations, and the non-relativistic scope condition.*

10 Derivation of F5 (Compatibility with U4)

This is the load-bearing derivation. It establishes $\hat{H}_{U3} = \hat{H}_{U4}$ via Galilean Lie algebra closure (Strategy A), with Stone-theorem uniqueness (Strategy B) as cross-check.

10.1 The exact compatibility condition

After F1–F4, two self-adjoint operators are in play on $\mathcal{H}$:

- $\hat{H}_{U3}$: the unique self-adjoint generator of $\{U_t\}$, abstractly the "time-translation generator."
- $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$: derived in U4 via Galilean integration on the U5 momentum operator.

The F5 question:

$$\hat{H}_{U3} \stackrel{?}{=} \hat{H}_{U4} \quad \text{as self-adjoint operators on } \mathcal{H}. \tag{4}$$

10.2 Discipline: identification, not derivation

Per §3.4, the F5 question is not "derive $\hat{H}_{U3}$ from $\hat{H}_{U4}$" (which would close the U3 $\leftrightarrow$ U4 circular loop). The question is whether the two operators, each constructed independently, are the same operator on $\mathcal{H}$. The protocol:

- $\hat{H}_{U3}$ is derived from F1 + F2 (Stone's theorem on time translations). U4 plays no role.
- $\hat{H}_{U4}$ is treated as a mathematical object on $\mathcal{H}$. U4's derivation chain is taken as background.
- F5 establishes an identification via shared structural infrastructure (the Galilean Lie algebra), not via deriving one from the other.

10.3 The Galilean Lie algebra at primitive level

The Galilean Lie algebra is generated by $\hat{H}$ (time translations), $\hat{p}_i$ (spatial translations; the U5 momentum operator), $\hat{x}_i$ (position), $\hat{K}_i$ (Galilean boosts), and $\hat{J}_i$ (rotations). The substantive commutators are:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\,\delta_{ij}, \tag{5}$$

$$[\hat{K}_i, \hat{p}_j] = i\hbar m\,\delta_{ij}, \tag{6}$$

$$[\hat{H}, \hat{K}_i] = -i\hbar\,\hat{p}_i. \tag{7}$$

The mass m enters (6) as the Galilean group's central-extension structure constant (Bargmann 1954) [10]. The boost generator takes the form

$$\hat{K}_i = m\,\hat{x}_i - t\,\hat{p}_i, \tag{8}$$

forced by the requirement that $\hat{K}_i$ generate the standard Galilean boost $\hat{p} \to \hat{p} - mv$.

The Galilean Lie algebra is *primitive-level non-relativistic-scope infrastructure*, not specific to U4's derivation. U4 *uses* the algebra to derive $\hat{H}_{U4}$'s form; U3 *uses* the same algebra (independently) to identify $\hat{H}_{U3}$ as the time-translation generator within the algebra's representation.

10.4 The U2 Galilean representation

All Galilean transformations act unitarily on $\mathcal{H}$:

- Spatial translations $T_a = \exp(i\hat{p} \cdot a/\hbar)$: unitary by U5's F1.
- Time translations $U_t = \exp(-i\hat{H}_{U3}t/\hbar)$: unitary by U3's F1.
- Rotations R_ω: unitary by primitive-level rotation invariance.
- Galilean boosts $B_v = \exp(i\hat{K} \cdot v/\hbar)$: unitary as exponential of self-adjoint $\hat{K}$.

The U2 Hilbert space carries a unitary representation of the Galilean group.

10.5 Strategy A — Galilean Lie algebra closure

By F1 + F2, $\hat{H}_{U3}$ is the unique self-adjoint generator of $\{U_t\}$ on $\mathcal{H}$. By §10.4, $\{U_t\}$ is part of the Galilean group's unitary representation on $\mathcal{H}$. Therefore $\hat{H}_{U3}$ is **the time-translation generator within the Galilean representation** — it plays the role of $\hat{H}$ in (5)–(7). In particular, $\hat{H}_{U3}$ must satisfy (7):

$$[\hat{H}_{U3}, \hat{K}_i] = -i\hbar\,\hat{p}_i. \tag{9}$$

This is an algebraic constraint inherited from the Galilean group's structure — not an assumption imported from U4.

Substituting (8) into (9), and using $[\hat{H}_{U3}, t\hat{p}_i] = 0$:

$$m\,[\hat{H}_{U3}, \hat{x}_i] = -i\hbar\,\hat{p}_i. \tag{10}$$

Decomposing $\hat{H}_{U3} = T_{U3}(\hat{p}) + V_{U3}(\hat{x})$ (a working ansatz justified independently in §10.6) and using $[T_{U3}(\hat{p}), \hat{x}_i] = -i\hbar\,\partial T_{U3}/\partial\hat{p}_i$:

$$m \cdot (-i\hbar)\,\frac{\partial T_{U3}(\hat{p})}{\partial \hat{p}_i} = -i\hbar\,\hat{p}_i,$$

so

$$\frac{\partial T_{U3}(\hat{p})}{\partial \hat{p}_i} = \frac{\hat{p}_i}{m}. \tag{11}$$

This is the same differential equation U4 derived. Integrating with $T_{U3}(0) = 0$:

$$T_{U3}(\hat{p}) = \frac{|\hat{p}|^2}{2m}. \tag{12}$$

The factor of 1/2 emerges from the chain-rule integration Jacobian, identical to U4's §F4 derivation.

10.6 Independent justification of the kinetic-plus-potential decomposition

The decomposition $\hat{H}_{U3} = T_{U3}(\hat{p}) + V_{U3}(\hat{x})$ is justified at the F5 level by primitive-level translation invariance + locality (sourced to the same primitives U4 used, but invoked independently here):

- **Translation invariance** of the participation graph (Primitives 03 + 06) forbids the free part of $\hat{H}_{U3}$ from depending on $\hat{x}$. The free part is $T_{U3}(\hat{p})$.
- **Locality** of external potentials forbids the potential part from depending on $\hat{p}$. The potential part is $V_{U3}(\hat{x})$.
- **Cross terms** are forbidden in the gauge-coupling-free scope.

The argument is primitive-level, sourced independently of U4's derivation chain.

10.7 The identification

Combining (12) with §10.6 and restoring $\hbar$:

$$\hat{H}_{U3} = \frac{\hbar^2|\hat{p}|^2}{2m} + V_{U3}(\hat{x}). \tag{13}$$

Comparing with U4:

$$\hat{H}_{U4} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x}). \tag{14}$$

The kinetic terms are identical. The potential terms agree as $V_{U3}(\hat{x}) = V(\hat{x})$ (both denote the same external-context-supplied scalar function of position; tautological at the F5 level).

The additive vacuum-energy zero-point gauge: Stone's theorem leaves $\hat{H}_{U3}$ unique up to an additive real constant; U4's $V(\hat{x})$ has the same additive-constant ambiguity. Both are the same gauge and coincide trivially.

Therefore:

$$\boxed{\hat{H}_{U3} = \hat{H}_{U4} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x}).} \tag{15}$$

The mass m is inherited per Arc M's "form FORCED, value INHERITED" methodology; $\hbar$ is inherited per the dimensional-atlas Madelung anchoring. **Strategy A closes the F5 identification. Falsifier Fal-5 dispatched.**

10.8 Strategy B (cross-check) — Stone-theorem uniqueness on $\hat{H}_{U4}$-generated group

The operator $\hat{H}_{U4}$ is self-adjoint on a dense domain in $\mathcal{H}$. By the spectral theorem, $\hat{H}_{U4}$ generates a strongly continuous one-parameter unitary group:

$$V_t := \exp(-i\hat{H}_{U4}t/\hbar). \tag{16}$$

The group $\{V_t\}$ acts on $\mathcal{H}$ as a unitary representation of the time-translation symmetry. By the uniqueness of unitary representations of $(\mathbb{R}_t, +)$ within the Galilean group's representation on $\mathcal{H}$ (sourced to Galilean group structure), $\{V_t\}$ coincides with $\{U_t\}$ from F1.

Stone's theorem guarantees a unique self-adjoint generator of the common group $\{V_t\} = \{U_t\}$. The unique generator is $\hat{H}_{U4}$ (by construction in (16)) and $\hat{H}_{U3}$ (by F2). Therefore $\hat{H}_{U4} = \hat{H}_{U3}$.

Strategy B confirms Strategy A's identification by an independent route.

Theorem 10.1 (F5 — Compatibility with U4)**.** *Let $\hat{H}_{U3}$ denote the unique self-adjoint generator of the time-translation group $\{U_t\}$ on the U2 Hilbert space $\mathcal{H}$, supplied by Stone's theorem applied in F1 + F2. Let $\hat{H}_{U4} = \hbar^2|\hat{p}|^2/(2m) + V(\hat{x})$ denote the kinetic-plus-potential operator derived in U4. Then, modulo the additive vacuum-energy zero-point gauge,*

$$\hat{H}_{U3} = \hat{H}_{U4} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x})$$

as self-adjoint operators on $\mathcal{H}$. The identification is established by Strategy A (Galilean Lie algebra closure) and cross-checked by Strategy B (Stone-uniqueness).

10.9 Schrödinger evolution forced

Combining F1, F2, F3, F4, F5:

$$\boxed{i\hbar\,\partial_t P(t) = \left(\frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x})\right) P(t).} \tag{17}$$

This is the Schrödinger equation in standard form, derived structurally with no PDE-level Schrödinger assumption, no U4-as-derivation-input circularity, no relativistic kinematics, and no downstream gauge-coupling structure.

11 Falsifier Resolution Table

Falsifier	Target	Dispatch argument	Section
Fal-1	F1	$\langle U_t P \mid U_t Q\rangle = \langle P \mid Q\rangle$ via translation-invariance of U2 measure on time axis	§6.4
Fal-2	F2	Stone's theorem on $\{U_t\}$ produces a unique self-adjoint generator	§7.2
Fal-3	F3	Linear unitary representations induce linear evolution equations	§8.2
Fal-4	F4	Operator-exponential differentiation gives first-order; non-rel scope selects abelian $\mathbb{R}_t$ over Lorentz boosts	§9.2
Fal-5	F5	Strategy A (Galilean Lie algebra closure) + Strategy B (Stone-uniqueness on $\hat{H}_{U4}$-generated group)	§10
Fal-6	F2	Stone's theorem produces a self-adjoint generator from a strongly continuous unitary group	§7.3
Fal-7	F1	Pre-derivation Primitive 13 audit confirms continuous time at Phase-1 scope; L^2-translation continuity inherits	§6.6

All seven falsifiers dispatched.

12 Conditionality Ledger

The U3 verdict is **forced** with the following inherited conditionalities and **zero new ones**:

1. Non-relativistic single-particle scope.
2. Gauge-coupling-free scope (inherited from U4).
3. U2 inner-product structure (both regimes).

4. U1 participation-measure structure.
5. U5 momentum-operator structure.
6. U4 Hamiltonian form (as identification target only).
7. Inherited values: mass m per Arc M; $\hbar$ per dimensional-atlas Madelung anchoring.
8. Primitive 13 continuous-time interpretation.
9. Galilean-group uniqueness within non-relativistic scope.

U3 introduces NO new structural candidates. All nine conditionalities are inherited from prior structural commitments.

13 Final U3 Theorem

Theorem 13.1 (U3 — Time-Translation Generator and Schrödinger Evolution)**.** *Let the ED primitive stack supply: graph homogeneity (Primitive 03), bandwidth (Primitive 04), spatial gradient (Primitive 06), tension polarity (Primitive 09), and relational timing (Primitive 13). Let the participation-measure carrier (U1, FORCED), the U2 inner product (U2-Discrete + U2-Continuum, both FORCED), the momentum operator (U5, FORCED), and the kinetic-plus-potential Hamiltonian operator (U4, FORCED-conditional) supply the Hilbert space $\mathcal{H}$ and operator infrastructure. Then:*

1. ***Time-translation representation.*** *The time-translation operators $(U_t P)_K(x,s) := P_K(x, s-t)$ for $t \in \mathbb{R}$ form a strongly continuous one-parameter abelian unitary group on $\mathcal{H}$.*
2. ***Unique self-adjoint generator.*** *By Stone's theorem, there exists a unique self-adjoint operator $\hat{H}_{U3}$ on $\mathcal{H}$ such that*
$$U_t = \exp\left(-\frac{i\hat{H}_{U3}t}{\hbar}\right).$$
3. ***Linear first-order-in-time evolution equation.*** *Differentiating yields $i\hbar\,\partial_t P(t) = \hat{H}_{U3}P(t)$, linear in $P(t)$, first-order in ∂_t.*
4. ***Compatibility with U4.*** *The Galilean Lie algebra forces $\hat{H}_{U3}$ to satisfy $[\hat{H}_{U3}, \hat{K}_i] = -i\hbar\hat{p}_i$ with the boost generator $\hat{K}_i = m\hat{x}_i - t\hat{p}_i$. Integration against the U5 momentum operator yields $T(\hat{p}) = |\hat{p}|^2/(2m)$ via the chain-rule Jacobian. The kinetic-plus-potential decomposition follows from translation invariance + locality. Modulo the additive vacuum-energy zero-point gauge,*
$$\hat{H}_{U3} = \hat{H}_{U4} = \frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x}).$$
5. ***Schrödinger evolution.*** *Combining (1)–(4):*
$$i\hbar\,\partial_t P(t) = \left(\frac{\hbar^2|\hat{p}|^2}{2m} + V(\hat{x})\right)P(t).$$

The derivation invokes no PDE-level Schrödinger assumption, no U4-as-derivation-input circularity, no relativistic kinematics, and no downstream gauge-coupling structure. All seven falsifiers are dispatched.

Status: FORCED-unconditional in both discrete and continuum regimes; gauge-invariant under the U2-Continuum conformal redundancy; **U4 promoted retroactively to forced-unconditional**; Phase-1 program structurally complete modulo description-level continuum gauge.

14 Downstream Consequences

14.1 U4 retroactive promotion

U4 was closed FORCED-conditional on U3 via Framing 1. With U3 now FORCED, **U4 promotes from forced-conditional to forced-unconditional retroactively**, resolving its sibling-CANDIDATE conditionality.

14.2 Schrödinger evolution forced-unconditional

The Phase-1 Step 2 chain reads, post-U3:

```
Primitives 03, 04, 06, 07, 09, 13
              |
              v
        U1 (FORCED) -> P_K = sqrt(b) * exp(i pi)
              |
              v
        U2 (FORCED) -> <P|Q> inner product
              |
              v
        U5 (FORCED) -> spatial translation -> p_hat = -i hbar nabla
              |
              v
        U4 (FORCED-UNCONDITIONAL post-U3)
              |
              v
        U3 (FORCED, this paper)
              |
              v
       SCHRÖDINGER EVOLUTION (FORCED-unconditional)
```

14.3 Updated structural-foundations ledger

#	Theorem	Status (post-U3)
10	Born rule (Gleason–Busch)	FORCED-unconditional
11	U2-Discrete	FORCED-unconditional
12	U2-Continuum (with conformal gauge)	FORCED-unconditional
13	U5 (translation generator + momentum operator)	FORCED-unconditional
14	U1 (participation-measure construction)	FORCED-unconditional
15	U4 (specific Hamiltonian form)	**forced-unconditional** (retroactive)
16	**U3 (this paper)**	**forced-unconditional**

Forced-theorem count: 15 → 16, with U4 retroactively promoted.

14.4 Active CANDIDATE inventory

Pre-U3: $\{U3\}$ + continuum gauge convention. Post-U3: $\{\,\}$ + continuum gauge convention. **The active upstream-candidate inventory is empty.**

14.5 All four foundational QM postulates structurally derived

Postulate	Pre-U3	Post-U3
Born rule	FORCED-unconditional	(unchanged)
Bell–Tsirelson bound	FORCED-unconditional	(unchanged)
Heisenberg uncertainty	FORCED-unconditional	(unchanged)
Schrödinger evolution	FORCED-conditional on U3	**forced-unconditional**

All four foundational postulates of non-relativistic single-particle quantum mechanics are now forced-unconditional structural consequences of the ED primitive ontology.

14.6 Phase-1 program structurally complete

The QM-emergence Phase-1 structural-foundations program is **structurally complete** modulo the description-level continuum gauge convention. The seven-arc cycle (born_gleason → U2 → U2-continuum → U5 → U1 → U4 → U3) closes the program.

15 Discussion and Outlook

15.1 The kinematics-dynamics seam closed

The structural-foundations cycle decomposes into three layers:

- **Layer 1 (carrier):** U1.
- **Layer 2 (Hilbert-space arena):** U2-Discrete + U2-Continuum + Theorem 10 + U5.
- **Layer 3 (dynamical):** U4 + U3.

U5 supplies the spatial-translation generator (kinematic backbone); U3 supplies the time-translation generator (dynamical backbone); U4 supplies the form of the time-translation generator. With U3 closed, the seam between kinematics and dynamics is structurally closed: the same Stone's-theorem skeleton applies on both axes; the Galilean Lie algebra closes the loop, identifying the two generators with their U4-derived operator content.

The Schrödinger equation is, at root, **what falls out when Stone's theorem is applied to time-translation symmetry on the participation-measure Hilbert space, and the resulting generator is identified with the Galilean-Lie-algebra-forced kinetic-plus-potential operator.**

15.2 Methodological completeness across seven arcs

The structural-derivation methodology has been demonstrated against seven substantively different structural questions:

1. **born_gleason** — Born rule via Gleason–Busch + non-contextuality.
2. **U2-Discrete** — sesquilinear inner product via primitive-level aggregation.
3. **U2-Continuum** — discrete-to-continuum lift with explicit conformal gauge.
4. **U5** — Stone's theorem on spatial translations → momentum operator.
5. **U1** — participation-measure construction via Frobenius + Cauchy functional equation.
6. **U4** — Galilean Lie algebra integration → kinetic-energy form, factor-of-2 Jacobian.

7. **U3 (this paper)** — Stone's theorem on time translations + Galilean closure → Hamiltonian operator + Schrödinger evolution.

The pattern (decompose CANDIDATE → identify load-bearing items → close via primitive-level arguments → introduce zero new CANDIDATEs) holds across all seven arcs.

15.3 What this changes

Standard quantum mechanics adopts the Schrödinger equation as a postulate. The U3 result establishes that this postulate is structurally derivable: time-translation symmetry (Primitive 13) plus the U2 Hilbert space plus Stone's theorem produces the equation's existence + uniqueness + linearity + first-order character; the Galilean Lie algebra plus the U5 momentum operator forces the generator's specific kinetic-plus-potential form. No PDE-level Schrödinger assumption is required.

The same structural depth now applies to all four foundational QM postulates. The Hilbert-space postulate, the Born-rule postulate, the position-momentum-conjugacy postulate, and the Schrödinger-evolution postulate are no longer separate axioms floating above the framework — they are derived structural consequences of the participation-graph ontology plus standard mathematical infrastructure.

15.4 What this does not change

The U3 result does not predict new experiments. The Schrödinger equation is the same equation it has always been; Event Density reproduces standard non-relativistic single-particle quantum mechanics exactly. What changes is the depth at which the answer to "why does quantum mechanics look like this?" can be given: in the standard formulation, "because we postulate it"; in Event Density, "because the participation-graph primitives plus standard mathematics force it."

15.5 Next steps

1. **Synthesis paper revision.** The QM-emergence Phase-1 synthesis paper requires comprehensive revision to reflect the post-U3 state.

2. **Phase-3 (gravitational sector) extensions.** With Phase-1 closed, the natural follow-on is the Phase-3 gravitational-sector program.

3. **Companion public-facing explainer.** The capstone explainer of the public-facing series, announcing Phase-1 closure to the general scientific audience.

4. **Empirical-test program.** With the structural-derivation program closed, the empirical-test program (FRAP, optomechanics, BEC, retrodictions) becomes the load-bearing locus of the framework's continued development.

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Manuscript prepared April 2026. Source materials: `arcs/U3/` four-memo arc and closure memo, formalized into a single publishable document. The U3 arc is the seventh and final of the structural-foundations cycle; this paper closes the QM-emergence Phase-1 structural-foundations program at the publication-record level.
Draft complete. Pausing for revision.